



Winding Tree

Lif Token State Channels



Augusto Lemble

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169



Scalability is one of the main concerns not just for Winding Tree, but for all blockchain projects. The developer community already offers a few potential solutions to this problem, like Plasma or Polka Dot, but the most promising concept, in my opinion, is “state channels”.

After a week of reading, playing and testing different state channels implementations, I managed to create a simple ERC20 contract that allows holders of Líf to initiate state channels between two parties (two, for now, baby steps!).

What is a state channel anyway?

A state channel is a way for two (or more) parties to transact off-chain. Once in a while, they reconcile all the off-chain transactions (“txs”) with the main net, paying just for one tx instead of paying for each one.

But Why?

Because interacting with the main net is expensive. Also, state channels have a bonus of not exposing all the private information that your txs could

contain.

How does it work?

Every time you sent crypto to someone else you have to sign that transaction with your private key, it's a way for the network to identify you. With state channels, you can share one signature that will be valid not just for one, but for several transactions with a group of people, instead of one person.

Why is the smart contract needed?

The smart contracts that, in this particular case, lives on the Ethereum blockchain, allows for parties to open and close state channels and also resolve disputes in an automated way.

It's just an example of how we will use state channels on the Winding Tree platform, and there's a lot more work to do. Check our GitHub repository to see the code and try and run it on your machine.

Special thanks to OpenZeppelin team for providing and maintaining the ECRcovery lib (used in the LiFChannels contract) and many other tools that help the whole community in building robust blockchain applications.

You can see our LifChannels contract [here](#)

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	verification				contract configuration - Approve minting of new LIFs to the "customers rewards" fund	
Registration / Invalidation	-Registers itself -Deposits a proper amount of LIF -Sends a LIF deposit withdrawal request (become disabled) -Disable itself -Become disabled if his LIF deposit balance is exhausted	-Registers itself -Deposits a proper amount of LIF -Sends a LIF deposit withdrawal request (become disabled) -Disable itself -Become disabled if his LIF deposit balance is exhausted	-Registers itself -Deposits a proper amount of LIF -Sends a LIF deposit withdrawal request (become disabled) -Disable itself -Become disabled if his LIF deposit balance is exhausted			
Verification		-Verifies suppliers or retailers -Invalidates suppliers or retailers -Pays verification fee with LIF to the DAO from its deposit balance			-Validates and verifies verifiers -Invalidates verifiers	
Deals creation			-Creates a deal on behalf of a customer and pays an offer price	-Creates a deal and pays an offer price		
Deals rejection	-Pays tx expenses -Pays a proper amount of LIF from its deposit as a rejection penalty			-Receives a full refund of the deal funds	-Receives "rejection fee" from the supplier	
Deals cancellation	-Receives full part of	-Receives part of the	-Receives part of the	-Pays tx expenses	-Receives part of the	

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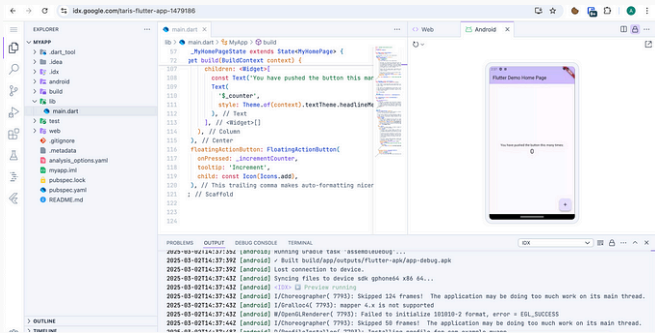
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